CHUNG-YEH(OLIVER) YANG

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EDUCATION

Columbia University New York, NY
M.S. in Data Science, GPA: 4.0/4.0 Dec 2026 (Expected)
Relevant Coursework: Agentic AI, Computer System, Computer Vision, Exploratory Data Analysis, Operational Logistics, Machine Learning, Poisson Process

Boston University Boston, MA
B.S. in Data Science, GPA: 3.8/4.0 Sep 2020 - Dec 2024
Major in Data Science, Minor in Business Administration
Coursework: Bayesian Statistics, Computer Systems, Database Design, Data Structure Algorithms, Deep Learning, Hypothesis Testing, Linear Algebra, Machine Learning, Multivariable Calculus

WORK EXPERIENCE

BCGX New York, NY
Incoming Summer Forward Deployed AI Scientist June 2026 – August 2026
- Developed data and modeling components for a **network optimization** system for a national vehicle retailer, projecting a **15%** OTP improvement and **20%** cost reduction in backtesting
- Built a **Python/SQL** pipeline on **Snowflake** to clean and validate route, volume, and cost data, forming the team's core dataset
- Developed cost, time, and volume sub-models feeding a **Gurobi MILP** location optimizer for hub and lane routing decisions

PalAI New York, NY
AI/ML Intern Sep 2025 – May 2026
- Designed a data pipeline to standardize and augment 100+ essay/feedback pairs, **fine-tuning** the model with the **OpenAI API**
- Implemented **RAG pipeline** with parallel vector search across 4 embedding sources using **pgvector**
- Built multi-agent backend with **tool-calling** LLM loop, dispatching 6 Supabase tools for real-time student data retrieval

Boston University Questrom School of Business Boston, MA
Research Assistant in Applied AI in Business Aug 2023 - Jan 2025
- Collaborated with **5+ industry researchers** to design and implement four AI solutions tailored to specific business challenges
- Analyzed over 50,000 user queries to measure the impact of query characteristics on **RAG generation** and user behaviors
- Engineered a pipeline that uses an **LLM** to label artwork genres from unsupervised clusters, achieving **zero-shot topic modeling**

BU Spark! Boston, MA
Data Science Technical Project Manager Sep 2024 - Dec 2024
- Managed six data analysis and visualization projects, coordinating with clients and facilitating student communication
- Provided technical assistance and leadership on data visualization projects leveraging **Python, Tableau,** and **Google BigQuery**
- Configured dataset and software setup for Notion and **GitHub**, ensuring teams had resources for project milestones

PROJECTS

Voyager – Travel Itinerary Optimizing Agent Jan 2026– May 2026
- Architected a 7-agent travel planner (**OpenAI Agents SDK**) that converts NL prompt into bookable multi-city itineraries
- Designed **Pydantic**-typed tools across APIs and a replanner that re-optimizes on disruption while preserving locked bookings
- Validated on a golden-eval set spanning multi-city, weather-disrupted, and budget-constrained scenarios with 100% pass rate

Text-to-SQL Data Intelligence Agent (Capstone Project) Jan 2026 – May 2026
- Built a production-grade text-to-SQL chatbot using **LangGraph**, enabling business users to query databases conversationally
- Orchestrated a multi-agent pipeline, including intent router, SQL generation, and **ECharts** visualization for end-to-end analytics
- Engineered a self-evaluation retry-loop that lifted complex-query success from **70% to 100%** within three retry iterations

YouBike Prediction Service Jun 2025 – Apr 2026
- Developed a serverless data pipeline on **AWS (Lambda, S3)** to collect and process real-time YouBike API data
- Engineered time-series features to train a **LightGBM** model, achieving over 90% recall in predicting critical station states
- Deployed and dockerized and prediction pipeline using AWS Lambda and ECR for automated, live inference

SKILLS

Programming Languages & Data Tools: Python, SQL, FastAPI, Apache Echarts, R, Redis, Tableau
AI/ML Frameworks: Pydantic, PyTorch, Multi-Agent Orchestration, OpenAI API, MCP/tool-calling, RAG, LangChain
Cloud & DevOps: AWS (Lambda, S3, ECR), Docker, Git, Google Cloud Platform (GCP), Supabase